Physics Verification Document

Equations to Verify for the Fiber Raman Suppression Project

For the physics review

April 10, 2026

How to read this document

Each section states a physics equation we use, cites where it comes from in the standard literature, and points to the exact file and line number where it is implemented in code. The intent is to let you verify, by eye, that the code and the physics agree. Only physics equations are included here — optimization and machine-learning material is in a separate document.

Contents

1 Notation and Conventions **1**

2 The Generalized Nonlinear Schrödinger Equation **2**
- 2.1 Lab-frame form . . . 2
- 2.2 Interaction-picture form (what the code actually integrates) . . . 2
- 2.3 Code implementation . . . 3

3 Dispersion Operator **4**

4 Raman Response Function **5**

5 Initial Pulse Shape **6**

6 Cost Functional (Physical Interpretation) **7**

7 Forward Propagation Solver **8**

8 Time Window Sizing **9**

9 Adjoint Sensitivity Equation **10**
- 9.1 Derivation (sketch) . . . 11
- 9.2 Gradient with respect to the input phase . . . 11
- 9.3 Code implementation . . . 11

10 Verification Tests **12**
- 10.1 VERIF-01: Soliton shape preservation . . . 12
- 10.2 VERIF-02: Photon number conservation . . . 12
- 10.3 VERIF-03: Taylor remainder for the adjoint gradient . . . 13

10.4 VERIF-04: Cost function cross-check 13

11 Summary of Physics Quantities to Verify **14**

12 What the Tests Do and Do Not Prove **14**

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1 Notation and Conventions

- $u(z,t)$: slowly-varying electric field envelope in the co-moving frame, units $\sqrt{\mathrm{W}}$. Depends on propagation distance z and retarded time t.
- $\hat{u}(z,\omega)$: Fourier transform of $u(z,t)$ in time, units $\sqrt{\mathrm{W}}\cdot\mathrm{s}$.
- $\tilde{u}(z,\omega)$: interaction-picture field, related to $\hat{u}$ by a dispersive phase (§2.2).
- $\omega_0 = 2\pi c/\lambda_0$: carrier angular frequency. For $\lambda_0 = 1550\,\mathrm{nm}$, $\omega_0 \approx 1.216\times 10^{15}\,\mathrm{rad/s}$.
- $\beta_n = \mathrm{d}^n\beta\,\omega|_{\omega_0}$: n-th order dispersion coefficient of the propagation constant $\beta(\omega)$ about ω_0.
- γ: effective nonlinear coefficient, units $\mathrm{W}^{-1}\mathrm{m}^{-1}$.
- f_R: fractional Raman contribution (dimensionless), $f_R = 0.18$ for silica.
- $h_R(t)$: Raman response function, units s^{-1}, causal ($h_R(t) = 0$ for $t < 0$).
- $\mathcal{F}$, $\mathcal{F}^{-1}$: forward and inverse Fourier transforms in time.
- $*$: temporal convolution.

Code convention: the code uses an inverted FFT sign convention. Specifically $\hat{u} = \mathcal{F}^{-1}[u]$, so $u = \mathcal{F}[\hat{u}]$ in the code. This does not change any physical equation — it only affects where you write `fft` vs. `ifft`. Throughout this document, physics equations are written in the standard convention; the code-convention note is only relevant when comparing the two.

2 The Generalized Nonlinear Schrödinger Equation

2.1 Lab-frame form

The field envelope $u(z,t)$ propagates through a lossless single-mode fiber according to the generalized nonlinear Schrödinger equation (GNLSE) with dispersion, Kerr nonlinearity, Raman response, and self-steepening. In the frequency domain it reads:

$$\frac{\partial\hat{u}(z,\omega)}{\partial z} = iD(\omega)\,\hat{u}(z,\omega) + i\gamma\,\frac{\omega}{\omega_0}\,\mathcal{F}\big[(1-f_R)\,|u|^2\,u \;+\; f_R\left(h_R * |u|^2\right)u\big]\,. \tag{1}$$

This is Agrawal, *Nonlinear Fiber Optics* 6th ed., Eq. (2.3.36), specialized to the single-mode case. The left-hand side is the rate of change of the complex spectrum along the fiber. The right-hand side has two pieces:

1. **Linear dispersion** $iD(\omega)\hat{u}$, where

$$D(\omega) \;=\; \sum_{n\geq 2}\frac{\beta_n}{n!}\,(\omega-\omega_0)^n, \tag{2}$$

excluding β_0 (absolute phase, unphysical) and β_1 (group velocity, absorbed into the co-moving frame).

2. **Nonlinear response** with two sub-pieces: an instantaneous Kerr term $(1 - f_R)\,|u|^2 u$ and a delayed Raman term $f_R\,(h_R * |u|^2)\,u$, both scaled by γ and by the self-steepening factor ω/ω_0.

Please verify

(a) The dispersion operator excludes β_0 and β_1.
(b) The nonlinear response is Kerr+Raman with $f_R = 0.18$ splitting.
(c) The self-steepening prefactor is ω/ω_0, not $(1 + \omega/\omega_0)$. This is equivalent to Agrawal's form under the carrier-absorbing transformation and is the convention used throughout the code.

2.2 Interaction-picture form (what the code actually integrates)

The dispersion term $iD(\omega)\hat{u}$ is stiff: it produces rapid phase oscillations that force the ODE solver to take tiny steps. We remove it analytically via the interaction-picture transformation:

$$\tilde{u}(z,\omega) \;=\; e^{-iD(\omega)\,z}\,\hat{u}(z,\omega). \tag{3}$$

Substituting into Eq. (1) and simplifying, the linear term cancels and we are left with the reduced ODE that the code integrates:

$$\boxed{\frac{\mathrm{d}\tilde{u}(z,\omega)}{\mathrm{d}z} \;=\; i\,e^{-iD(\omega)\,z}\,\frac{\omega}{\omega_0}\,\mathcal{F}\Big[\gamma\,(1-f_R)\,|u|^2 u \;+\; \gamma\,\big(h_R * |u|^2\big)\,u\Big]} \tag{4}$$

where $u(z,t)$ inside the brackets is the time-domain *lab-frame* field, reconstructed from $\tilde{u}$ via

$$u(z,t) \;=\; \mathcal{F}^{-1}\Big[e^{+iD(\omega)\,z}\,\tilde{u}(z,\omega)\Big]. \tag{5}$$

Why the interaction picture helps. The linear dispersion causes frequency components to acquire phase at the rate $D(\omega)$. For typical SMF-28 over 1 m of propagation, $D(\omega)L \sim 100$ radians at the edge of the signal band. A direct integrator would need thousands of sub-steps to resolve this phase. In the interaction picture, the linear phase is handled analytically by Eq. (5), and the solver only needs to resolve the much slower nonlinear dynamics — typically hundreds of steps per meter instead of millions.

2.3 Code implementation

The reduced ODE Eq. (4) is the right-hand side of the ODE solver. The function is `disp_mmf!` in `src/simulation/simulate_disp_mmf.jl`, lines 25–61.

Listing 1: `src/simulation/simulate_disp_mmf.jl` L25–L61

```
25 function disp_mmf!(d    ,      , p, z)
26     selfsteep, D  ,    , hR  , one_m_fR, attenuator, ..., = p
27
28     @. exp_D_p = cis(D   * z)                          # L28:  e^{+iDz}
29     @. exp_D_m = cis(-D   * z)                         # L29:  e^{-iDz}
30
31     @. u   = exp_D_p *                                   # L31:  lab frame (
       Eq. 4)
```

```
32
33      fft_plan_M! * u                                    # L33:  -> time domain
34      @. ut = attenuator * u                             # L34:  boundary window
35      @. v = real(ut)
36      @. w = imag(ut)
37
38      # Kerr:  _K  = (1-fR)    |u|^2 u
39      @tullio  Kt [t,i,j] =   [i,j,k,l]*(v[t,k]*v[t,l] + w[t,k]*w[t,l])   # L39
40      @tullio  K [t,i] =  Kt [t,i,j] * v[t,j]            # L40
41      @tullio  K [t,i] =  Kt [t,i,j] * w[t,j]            # L41
42      @.  Kt  =  K  + 1im *  K                           # L42
43      @.  Kt  *= one_m_fR                                # L43:  (1-fR) factor
44
45      # Raman:  _R  =    (h_R * |u|^2) u
46      @.  Kt_cplx  = ComplexF64( Kt , 0.0)               # L46
47      fft_plan_MM! *  Kt_cplx                            # L47:  -> freq for convolution
48      @. hR _ R  = hR  *  Kt_cplx                        # L48:  freq-domain convolution
49      ifft_plan_MM! * hR _ R                             # L49:  -> back to time
50      fftshift!(hR_conv_ R, hR _ R , 1)                  # L50
51      @.  Rt  = real(hR_conv_ R)
52      @tullio  R [t,i] =  Rt [t,i,j] * v[t,j]            # L52
53      @tullio  R [t,i] =  Rt [t,i,j] * w[t,j]            # L53
54      @.  Rt  =  R  + 1im *  R
55
56      @.  t  =  Kt  +  Rt                                # L57:  Kerr + Raman
57      ifft_plan_M! *  t                                  # L58:  -> freq
58       t  .*= selfsteep                                  # L59:    / factor
59      @. d    = 1im * exp_D_m *  t                       # L60:  i * e^{-iDz} * (...)
60  end
```

Line-by-line correspondence:

- 28–29: the two phase factors $e^{\pm iDz}$ from Eqs. (3), (5).
- 31: inverse interaction-picture transform, Eq. (5) in the frequency domain.
- 33: FFT to time to evaluate the nonlinear response pointwise.
- 39: Kerr operator $\delta_K = \gamma\,|u|^2$ via the contraction $\gamma_{ijkl}(v_k v_l + w_k w_l)$ with $v = \operatorname{Re} u$, $w = \operatorname{Im} u$.
- 43: factor $(1 - f_R)$ on the Kerr term (this is the $(1 - f_R)$ in Eq. (4)).
- 48: frequency-domain Raman convolution $\widehat{h_R} \cdot \widehat{|u|^2}$, which becomes $h_R * |u|^2$ in time.
- 59: self-steepening factor ω/ω_0 (the array `selfsteep` is built once at setup from ω_s/ω_0).
- 60: the overall factor $i\,e^{-iDz}$ of Eq. (4).

Please verify

Does the sequence of operations on lines 28–60 correctly assemble Eq. (4)? The key points are: (i) the lab-frame field on line 31 is exactly Eq. (5), (ii) the Kerr factor on line 43 is $(1 - f_R)$ with no additional scaling, (iii) the Raman convolution on line 48 is in frequency domain, (iv) the self-steepening on line 59 uses ω/ω_0, and (v) the final $i\,e^{-iDz}$ on line 60 closes the loop back to the interaction-picture derivative.

3 Dispersion Operator

The dispersion operator $D(\omega)$ is the Taylor expansion of the propagation constant $\beta(\omega)$ about ω_0, with β_0 and β_1 removed:

$$D(\omega) \;=\; \beta(\omega) - \beta(\omega_0) - \beta_1\,(\omega - \omega_0) \;=\; \sum_{n=2}^{N_\beta} \frac{\beta_n}{n!}\,(\omega - \omega_0)^n. \tag{6}$$

In the code, we set $\Omega \equiv \omega - \omega_0$ and work with the FFT-domain angular frequency grid $\Omega = 2\pi\,\Delta f$.

Fiber parameters used. The two presets are defined in `scripts/common.jl`, lines 47–76:

Preset	γ [$\mathrm{W^{-1}m^{-1}}$]	β_2, β_3 [SI]
`:SMF28`	1.1×10^{-3}	$(-2.17 \times 10^{-26},\ 1.2 \times 10^{-40})$
`:HNLF`	1.0×10^{-2}	$(-5.0 \times 10^{-27},\ 1.0 \times 10^{-40})$

Operator construction. The dispersion operator is built once at setup in `get_disp_fiber_params_user_defined`, `src/helpers/helpers.jl`, lines 161–193. Line 187 is the operative line:

Listing 2: `src/helpers/helpers.jl` L185–L188

```
185  _tail   = vcat(collect(betas_user), zeros( _order -1 - length(betas_user))
       )
186   n_      = reshape(vcat([0.0, 0.0],  _tail ), :, 1)        # [0, 0,      ,
        , ...]
187 D        = hcat([(2  *fftfreq(Nt, 1/ t )*1e12).^n / factorial(n)
188                    for n in 0:  _order ]...) *   n_
```

The two leading zeros are β_0 and β_1 respectively. The tail is $[\beta_2, \beta_3, \ldots]$. The Vandermonde-style matrix multiplication is exactly the sum in Eq. (6) evaluated at every grid point Ω_k.

Please verify

(a) The leading two entries of β`n_`ω are zero (co-moving frame, no absolute phase).
(b) The third entry is β_2 with the sign from Table above.
(c) The frequency grid used is $(\omega - \omega_0) = 2\pi\,\Delta f$, i.e. *offset* from the carrier.

4 Raman Response Function

The delayed Raman response is modeled by the Blow–Wood single-oscillator form (Agrawal Eq. 2.3.40):

$$h_R(t) \;=\; \frac{\tau_1^2 + \tau_2^2}{\tau_1\,\tau_2^2}\,e^{-t/\tau_2}\,\sin\!\left(\frac{t}{\tau_1}\right)\,\Theta(t), \tag{7}$$

where $\Theta(t)$ is the Heaviside step function (enforcing causality: the glass cannot respond before the excitation arrives), and the time constants for fused silica are

$$\tau_1 = 12.2\,\text{fs}, \qquad \tau_2 = 32\,\text{fs}. \tag{8}$$

The factor $(\tau_1^2 + \tau_2^2)/(\tau_1\tau_2^2)$ is a normalization such that $\int_0^\infty h_R(t)\,dt = 1$.

Code implementation. `src/helpers/helpers.jl`, lines 181–183 (for user-defined single-mode fibers; there is a structurally identical copy at L106–L108 for the GRIN path):

Listing 3: `src/helpers/helpers.jl` L181–L183

```
181 ts_pos = max.(ts, 0.0)
182 hRt = fR *  t  * 1e3 * ( 1 ^2 +  2 ^2) / ( 1  *  2 ^2) .*
183       exp.(-ts_pos * 1e15 /  2 ) .* sin.(ts_pos * 1e15 /  1 ) .*
184       (sign.(ts) .+ 1) / 2
185 hR   = fft(hRt)
```

Two things to notice:

- The factor `(sign(ts) + 1)/2` is the Heaviside step $\Theta(t)$. It equals 1 for $t > 0$, 0 for $t < 0$, and 1/2 at $t = 0$.
- The `max(ts, 0.0)` on line 181 is a numerical-stability guard, not physics. Without it, the `exp(-t/`τ_2`)` factor would overflow `Float64` for negative t (where $e^{|t|/\tau_2}$ diverges) even though the Heaviside would then multiply it by zero. Pre-clamping t to $\max(t, 0)$ makes the intermediate quantity finite. The result is mathematically identical to Eq. (7).

Please verify

(a) The Blow–Wood form matches your expectation for $h_R(t)$ in silica.
(b) The Heaviside factor is correctly present (causality).
(c) The $\max(t, 0)$ clamp is a numerical artifact only; it does not change any non-zero value of h_R.

5 Initial Pulse Shape

All simulations use a sech^2 pulse at $\lambda_0 = 1550\,\text{nm}$ with FWHM $= 185\,\text{fs}$. The time-domain envelope is

$$u(0,t) \;=\; \sqrt{P_{\text{peak}}}\,\text{sech}\left(\frac{t}{T_0}\right), \qquad T_0 = \frac{\text{FWHM}}{1.7627}, \tag{9}$$

where $1.7627 = 2\,\text{arcosh}\,\sqrt{2}$. The intensity $|u|^2 = P_{\text{peak}}\,\text{sech}^2(t/T_0)$ has the desired full-width at half-maximum.

The simulation works with average power P_{cont} (easier to specify experimentally). The conversion is

$$P_{\text{peak}} \;=\; 0.881374 \times \frac{P_{\text{cont}}}{\text{FWHM}\cdot f_{\text{rep}}}, \tag{10}$$

where $0.881374 = \int_{-\infty}^{\infty} \text{sech}^2(x)\,dx\,/\,(2\,\text{arcosh}\,\sqrt{2})$ is the sech^2 shape factor for converting peak-to-average.

Code implementation. src/simulation/simulate_disp_mmf.jl, lines 135–152:

Listing 4: src/simulation/simulate_disp_mmf.jl L142–L150

```
142 elseif pulse_form == "sech_sq"
143         = fwhm / 1.7627                                    # L143:  T_0
144     u0_norm .= sech.(-ts /   )                             # L144
145     P_peak = 0.881374 * P_cont / fwhm / rep_rate          # L145:  Eq.
        (10)
146 end
147 u0_norm .*= u0_modes'
148 u0_norm /=   maximum  (sum(abs2.(u0_norm), dims=2))          # L148:
    renormalize to peak = 1
149 ut0 = u0_norm *   P_peak                                     # L149:
    scale to   P_peak
150 u  0  = ifft(ut0, 1)                                        # L150:  to
    frequency domain
```

Soliton number. A key derived quantity is the soliton number

$$N = \sqrt{\frac{\gamma\, P_{\text{peak}}\, T_0^2}{|\beta_2|}}. \tag{11}$$

Across our sweeps, N ranges from ≈ 1.3 to ≈ 6.3, spanning the single-soliton and few-soliton regimes. This formula appears in scripts/common.jl line 457:

Listing 5: scripts/common.jl L454–L457

```
454  2   = betas_user[1]
455 T0 = pulse_fwhm / (2 * acosh(sqrt(2)))
456 P_peak = P_cont / (pulse_fwhm * pulse_rep_rate)
457 N_sol   = sqrt(gamma_user * P_peak * T0^2 / abs( 2 ))
```

Please verify

(a) The sech^2 envelope matches Eq. (9) including the $T_0 = \text{FWHM}/1.7627$ relation.
(b) The peak-to-average factor 0.881374 on line 145 matches your pulse-shape convention.
(c) The soliton number on line 457 matches the standard definition.

6 Cost Functional (Physical Interpretation)

The optimization minimizes the fractional spectral energy in a defined Raman band:

$$J = \frac{E_{\text{band}}}{E_{\text{total}}} = \frac{\displaystyle\int_{\text{band}} |\hat{u}(L,\omega)|^2\, d\omega}{\displaystyle\int_{-\infty}^{\infty} |\hat{u}(L,\omega)|^2\, d\omega} \in [0,1]. \tag{12}$$

The "band" is defined as all FFT bins with frequency offset $\Delta f < -13\,\text{THz}$ relative to the carrier, which captures the Stokes-shifted Raman content. The definition is the same for both fibers.

On the discrete grid, this becomes

$$J = \frac{\sum_{k\in\mathcal{B}} |\hat{u}_k|^2}{\sum_{k=1}^{N_t} |\hat{u}_k|^2}, \tag{13}$$

where $\mathcal{B}$ is the set of spectral bins in the Raman band.

Code implementation. `scripts/common.jl`, lines 260–276:

Listing 6: `scripts/common.jl` L260–L276

```
260 function spectral_band_cost(u_f, band_mask)
261     @assert size(u_f,1) == length(band_mask)
262     @assert any(band_mask)
263     @assert sum(abs2.(u_f)) > 0
264
265     E_band  = sum(abs2.(u_f[band_mask, :]))        # L266
266     E_total = sum(abs2.(u_f))                      # L267
267     J       = E_band / E_total                    # L268
268     dJ      = u_f .* (band_mask .- J) ./ E_total   # L269: terminal adjoint
269
270     @assert 0 ≤ J ≤ 1
271     @assert all(isfinite, dJ)
272     return J, dJ
273 end
```

The second return value `dJ` is the terminal condition for the adjoint sensitivity equation; it is the derivative of J with respect to $\hat{u}^*(L,\omega)$, which you can derive by differentiating Eq. (13).

Please verify

(a) The cost is energy-fractional and dimensionless, bounded in $[0,1]$.
(b) The Raman band threshold of $-13\,\text{THz}$ correctly captures the Stokes shift in silica.
(c) The gradient expression on line 269 is the correct derivative $\partial J/\partial \hat{u}^*(L)$.

7 Forward Propagation Solver

The ODE Eq. (4) is integrated from $z = 0$ to $z = L$ using Julia's `DifferentialEquations.jl`. We use `Tsit5` (Tsitouras' 5th-order Runge–Kutta method) with relative tolerance 10^{-8}.

`src/simulation/simulate_disp_mmf.jl`, lines 176–198:

Listing 7: `src/simulation/simulate_disp_mmf.jl` L176–L198

```
176 function solve_disp_mmf(u_0, fiber, sim)
177     p_disp_mmf = get_p_disp_mmf(..., sim["attenuator"])                # L177
178     prob_disp_mmf = ODEProblem(disp_mmf!, u_0, (0, fiber["L"]), p_disp_mmf)  # L179
179
180     if isnothing(fiber["zsave"])
181         sol = solve(prob_disp_mmf, Tsit5(), reltol=1e-8)               # L182
182         return Dict("ode_sol" => sol)
```

```
183     else
184         sol = solve(prob_disp_mmf, Tsit5(), reltol=1e-8,
185                     saveat=fiber["zsave"])                          #
                            L186
186         u_z = zeros(ComplexF64, length(fiber["zsave"]), sim["Nt"], sim["
            M"])
187         ut_z = zeros(ComplexF64, length(fiber["zsave"]), sim["Nt"], sim["M
            "])
188         for i in 1:length(fiber["zsave"])
189             u_z[i,:,:] = cis.(fiber["D  "]*fiber["zsave"][i]) .* sol(
                fiber["zsave"][i])
190             ut_z[i,:,:] = fft(u_z[i,:,:], 1)
191         end
192         return Dict("ode_sol"=>sol, "u_z"=>u_z, "ut_z"=>ut_z)
193     end
194 end
```

Line 192 converts the interaction-picture field $\tilde{u}$ back to the lab frame by multiplying by $e^{+iD(\omega)z}$, which is the inverse of the transformation in Eq. (3).

Boundary condition: super-Gaussian window. To prevent FFT wraparound, a super-Gaussian window of order 30 is applied to the time-domain field on every RHS evaluation (line 34 of `disp_mmf!`). This is a soft absorbing boundary condition — it attenuates field content that approaches the edge of the time window. The window is built in `get_disp_sim_params`, `src/helpers/helpers.jl` lines 59–63:

Listing 8: `src/helpers/helpers.jl` L59–L63

```
59 r_attenuation = 0.85 * time_window / 2          # half-width at onset
60 n_attenuation = 30                              # super-Gaussian order
61   _attenuation   = r_attenuation / log(2)^(1/n_attenuation)
62 ...
63 attenuator = exp.(-(abs.(1e12*ts)/  _attenuation  ).^n_attenuation) * ones(M
    )'
```

The functional form is

$$w(t) \;=\; \exp\left[-\left(\frac{|t|}{\sigma}\right)^{30}\right], \qquad \sigma \;=\; \frac{0.85\,W/2}{(\ln 2)^{1/30}}, \tag{14}$$

where W is the full time window. For $n = 30$, $w(t) \approx 1$ for $|t| < 0.85\,W/2$ and drops sharply to zero near $|t| = W/2$.

Please verify

(a) `Tsit5` with `reltol=1e-8` is an acceptable solver for this class of nonlinear Schrödinger equation at the soliton numbers we study ($N \leq 6$).
(b) The super-Gaussian boundary treatment does not interfere with the pulse physics as long as the time window is sized so that the pulse never reaches $|t| = 0.85\,W/2$. This is checked by the photon-conservation test (§10.2).

8 Time Window Sizing

The time window W must be large enough that the pulse does not reach the absorbing boundary at any point during propagation. Two effects cause the pulse to spread in time:

1. **Dispersive walk-off** between frequencies separated by the Raman shift $\Delta\omega_R = 2\pi \times 13\,\mathrm{THz}$. Different frequencies travel at different group velocities, so a pair of points separated by $\Delta\omega_R$ walk apart by
$$\Delta t_{\mathrm{walk}} \;=\; |\beta_2|\, L\, \Delta\omega_R. \tag{15}$$

2. **SPM-induced spectral broadening** converted back into temporal spread via GVD. The nonlinear phase accumulated over the fiber is
$$\phi_{\mathrm{NL}} \;=\; \gamma\, P_{\mathrm{peak}}\, L \qquad \text{(radians, dimensionless).} \tag{16}$$
The SPM-broadened angular bandwidth of a sech^2 pulse is (Agrawal Ch. 4)
$$\Delta\omega_{\mathrm{SPM}} \;=\; 0.86\,\frac{\phi_{\mathrm{NL}}}{T_0}. \tag{17}$$
Propagating this broadened spectrum through GVD produces an additional temporal spread of
$$\Delta t_{\mathrm{SPM}} \;=\; |\beta_2|\, L\, \Delta\omega_{\mathrm{SPM}}. \tag{18}$$

The recommended time window adds both terms with a safety factor:
$$W \;\gtrsim\; 2\,\big(\Delta t_{\mathrm{walk}} + \Delta t_{\mathrm{SPM}} + 0.5\,\mathrm{ps}\big). \tag{19}$$

Critical note — dimensional analysis. It is tempting to try to use ϕ_{NL} directly as a frequency scale, but ϕ_{NL} is *dimensionless* (radians). The division by T_0 in Eq. (17) is essential: $\mathrm{rad/s} = \mathrm{rad}\,/\,\mathrm{s}$. An earlier version of the code was missing this $1/T_0$ factor and produced time windows 5–7× too small at high power, which caused 38–49% photon number loss at the absorbing boundary. This was caught by the photon-conservation test (§10.2) and fixed.

Code implementation. `scripts/common.jl`, lines 191–215:

Listing 9: `scripts/common.jl` L191–L215

```
191 function recommended_time_window(L_fiber; safety_factor=2.0, beta2=20e-27,
192                                  gamma=0.0, P_peak=0.0, pulse_fwhm=185e
                                        -15)
193     @assert L_fiber > 0
194     @assert beta2 > 0
195     @assert pulse_fwhm > 0
196
197       _raman      = 2   * 13e12                                  # L198: 2
          13 THz
198     walk_off_ps = beta2 * L_fiber *    _raman    * 1e12       # L199: Eq.
         (19)
199     pulse_extent = 0.5
200
201     spm_ps = 0.0
```

```
202     if gamma > 0 && P_peak > 0
203         T0      = pulse_fwhm / 1.763                        # sech^2 half-duration [s]
204          _NL     = gamma * P_peak * L_fiber                  # L208: Eq. (20) [rad]
205           _SPM    = 0.86 *   _NL   / T0                        # L209: Eq. (21) [rad/s]
206         spm_ps = beta2 * L_fiber *    _SPM    * 1e12           # L210: Eq. (22) [ps]
207     end
208
209     total_ps = walk_off_ps + spm_ps + pulse_extent
210     return max(5, ceil(Int, total_ps * safety_factor))   # L214: Eq. (23)
211 end
```

Please verify

(a) The walk-off formula on line 199 corresponds to Eq. (15).
(b) Line 209 implements Eq. (17), including the dimensional $1/T_0$ factor.
(c) The safety factor of 2 in Eq. (19) is sufficient for your range of intended operating conditions.

9 Adjoint Sensitivity Equation

To optimize the input spectral phase efficiently, we use the adjoint method to compute the gradient $\partial J/\partial\varphi$ with only *one* additional backward propagation, instead of the N_t finite-difference perturbations that naïve differentiation would require.

9.1 Derivation (sketch)

Start from the optimization problem

$$\min_{\varphi(\omega)} J\big(\hat{u}(L)\big) \quad \text{subject to} \quad \frac{\partial \hat{u}}{\partial z} = F\big(\hat{u}, z\big), \quad \hat{u}(0,\omega) \;=\; \hat{u}_0(\omega)\, e^{i\varphi(\omega)}, \tag{20}$$

where F is the GNLSE right-hand side (Eq. (1)). Introduce a Lagrange multiplier field $\lambda(z,\omega)$ — the adjoint field — and write the Lagrangian

$$\mathcal{L} \;=\; J\big(\hat{u}(L)\big) \;+\; \int_0^L \mathrm{Re}\Big\langle \lambda,\, \frac{\partial \hat{u}}{\partial z} - F(\hat{u}, z)\Big\rangle\, dz. \tag{21}$$

Stationarity of $\mathcal{L}$ with respect to $\hat{u}(z)$ (after integration by parts in z) gives the **backward adjoint equation**

$$\boxed{-\frac{\mathrm{d}\lambda(z,\omega)}{\mathrm{d}z} \;=\; \left[\frac{\partial F}{\partial \hat{u}}\right]^{\dagger} \lambda(z,\omega)} \tag{22}$$

with the **terminal condition**

$$\lambda(L,\omega) \;=\; \left.\frac{\partial J}{\partial \hat{u}^*}\right|_{z=L}. \tag{23}$$

The † denotes the adjoint operator (conjugate transpose in the discretized problem). For our cost Eq. (13), the terminal condition evaluates to

$$\lambda(L,\omega) \;=\; \frac{\hat{u}(L,\omega)\big(\mathbb{K}_{\text{band}}(\omega) - J\big)}{E_{\text{total}}}, \tag{24}$$

which is exactly what `spectral_band_cost` returns as `dJ` on line 269 of `scripts/common.jl`.

9.2 Gradient with respect to the input phase

Once $\lambda(0,\omega)$ is known from backward integration, the gradient of J with respect to the input phase $\varphi(\omega)$ comes from the chain rule through $\hat{u}(0,\omega) = \hat{u}_0(\omega)\, e^{i\varphi}$:

$$\frac{\partial J}{\partial \varphi(\omega)} \;=\; 2\,\mathrm{Re}[\, i\,\hat{u}(0,\omega)\,\lambda^*(0,\omega)\,]\,. \tag{25}$$

9.3 Code implementation

The adjoint ODE right-hand side is `adjoint_disp_mmf!` in `src/simulation/sensitivity_disp_mmf.jl`, lines 32–79. It has the same overall structure as the forward solver but computes a linearized-transposed operator. The four physical pieces are:

- 72: Kerr+Raman δ_1 contraction ($|u|^2 u$ linearization)
- 73: conjugate Kerr δ_2 contraction (u^2 linearization)
- 74: forward-path Raman convolution with h_R
- 75: conjugate-path Raman convolution with h_R^*

which are summed on line 78 to produce the adjoint time-derivative. Each piece is the formal transpose of the corresponding forward term in `disp_mmf!`.

Numerical accuracy. The forward solver uses `Tsit5` with continuous interpolation. When the adjoint queries the forward field at an arbitrary z (via ~(z) on line 50), it is evaluated through `Tsit5`'s 4th-order Hermite interpolant. Consequently the adjoint's accuracy is bounded by that 4th-order interpolant, and using a higher-order backward solver would not improve anything. We use `Tsit5` with `reltol=1e-8` for the adjoint as well. This choice is validated by the Taylor-remainder test (§10.3).

Please verify

(a) The adjoint sensitivity equation Eq. (22) is the standard form for the backward sensitivity of an ODE-constrained optimization problem.
(b) The terminal condition Eq. (23) matches Eq. (24) for our cost functional.
(c) The phase-gradient formula Eq. (25) is the chain rule through the input phase mask.

10 Verification Tests

We run four independent physics tests at the production grid size $N_t = 2^{14}$ in `scripts/verification.jl`. Each is derived from a physical invariant or analytical solution that must hold if the code is correct.

10.1 VERIF-01: Soliton shape preservation

Physical principle. The fundamental $N = 1$ soliton is an exact analytical solution of the standard NLSE (Kerr nonlinearity with anomalous dispersion, no Raman):

$$u(z,t) \;=\; \sqrt{P_{\text{peak}}}\,\operatorname{sech}\!\left(\frac{t}{T_0}\right)\exp\!\left[\,i\,\frac{z}{2\,L_D}\,\right], \qquad P_{\text{peak}} \;=\; \frac{|\beta_2|}{\gamma\,T_0^2}. \tag{26}$$

After one soliton period $z_{\text{sol}} = (\pi/2)\,L_D$ (with $L_D = T_0^2/|\beta_2|$), the pulse returns to its original shape. Any deviation is numerical error.

Code implementation. `scripts/verification.jl` lines 73–150. The test runs at $\beta_2 = -2.6\times10^{-26}$ s^2/m, $\gamma = 1.3\times10^{-3}$ $\text{W}^{-1}\text{m}^{-1}$, $T_0 = 185\,\text{fs}/(2\,\text{arcosh}\,\sqrt{2})$, with f_R set to 10^{-15} (Raman effectively disabled).

Pass criterion: $\max|I_{\text{out}}(t)/\max I_{\text{out}} - I_{\text{in}}(t)/\max I_{\text{in}}| < 0.02$ (2%) across the pulse core (intensity >5% of peak).

Last result: max deviation = 0.0112 (**PASS**, 1.1% < 2%).

10.2 VERIF-02: Photon number conservation

Physical principle. In a lossless fiber, Raman scattering transfers *energy* from higher to lower frequencies by exciting phonons, so classical energy $\int|\hat{u}|^2\,d\omega$ is *not* conserved. But every Raman event takes one photon at frequency ω_1 and produces one photon at $\omega_2 < \omega_1$, so the *photon number*

$$N_{\text{ph}} \;=\; \int \frac{|\hat{u}(\omega)|^2}{\hbar\,\omega}\,d\omega \;\propto\; \int \frac{|\hat{u}(\omega)|^2}{\omega}\,d\omega \tag{27}$$

is conserved. This is the correct invariant to use for verifying a GNLSE with self-steepening and Raman (Agrawal Ch. 2.3; Brabec & Krausz, RMP 2000).

Code implementation. `scripts/verification.jl` lines 188–199 (`compute_photon_number`) and lines 211–252 (test loop over 5 production configurations). The $\hbar$ constant is dropped since we check the *ratio* $N_{\text{ph}}(L)/N_{\text{ph}}(0)$.

Pass criterion: $|N_{\text{ph}}(L)/N_{\text{ph}}(0) - 1| < 0.01$ (1%) for all 5 runs.

Last result (post SPM-formula fix): drifts < 1% on all 5 configurations (**PASS**). Prior to the fix, high-power runs showed up to 49% drift, which led directly to discovering and fixing the time-window dimensional error in §8.

Please verify

(a) Photon number, not energy, is the correct conservation law for the GNLSE with Raman (energy goes into phonons at each Raman event).
(b) The 1% tolerance is acceptable given the super-Gaussian boundary window, which is a physical approximation to an open boundary.
(c) The post-fix drifts are at the level of numerical solver tolerance plus attenuator leakage, not a physical error.

10.3 VERIF-03: Taylor remainder for the adjoint gradient

Physical principle. If ∇J is the correct gradient of J, then Taylor's theorem guarantees that

$$R(\epsilon) \;\equiv\; \big|\, J(\varphi + \epsilon\,\delta\varphi) - J(\varphi) - \epsilon\,\langle \nabla J,\, \delta\varphi \rangle \,\big| \;=\; O(\epsilon^2). \tag{28}$$

Plotting $\log R(\epsilon)$ versus $\log \epsilon$ gives a straight line of *slope* 2 if the gradient is correct. A slope of 1 signals a linear bias in the gradient (an error proportional to $\delta\varphi$); a slope of 0 signals a zero-order bias. Intermediate slopes near 2 (say, in $[1.7, 2.3]$) are treated as passing.

Code implementation. `scripts/verification.jl` lines 271–330. Runs at $L_{\text{fiber}} = 0.1$ m (short enough that higher-order Taylor terms are negligible). Computes R at $\epsilon \in \{1, 10^{-1}, 10^{-2}, 10^{-3}\}$.

Pass criterion: slopes between consecutive ϵ pairs all in $[1.4, 2.6]$.

Last result: slopes = [2.01, 2.07, 2.09], all in range (**PASS**). This is strong evidence that the hand-derived adjoint of §9 has no sign error, no missing factor, and no off-by-one in the chain rule.

10.4 VERIF-04: Cost function cross-check

Physical principle. The cost J is a simple ratio of integrals (Eq. (13)), so the result of the optimized routine `spectral_band_cost` must match a direct evaluation of $E_{\text{band}}/E_{\text{total}}$ to machine precision. This tests that the band-mask indexing is correct and that there are no `fftshift` confusions.

Last result: $|J_{\text{func}} - J_{\text{direct}}| = 0$ exactly (**PASS** at 10^{-12} tolerance).

11 Summary of Physics Quantities to Verify

Equation	Section	Code location
GNLSE, interaction picture, Eq. (4)	§2	`simulate_disp_mmf.jl` L25–61
Dispersion Taylor, Eq. (6)	§3	`helpers.jl` L185–187
Raman response, Eq. (7)	§4	`helpers.jl` L181–183
sech^2 pulse, Eq. (9)	§5	`simulate_disp_mmf.jl` L142–150
Peak power factor, Eq. (10)	§5	`simulate_disp_mmf.jl` L145
Soliton number, Eq. (11)	§5	`common.jl` L457
Cost functional, Eq. (13)	§6	`common.jl` L260–276
Time window (walk-off + SPM), Eq. (19)	§8	`common.jl` L191–215
Adjoint ODE, Eq. (22)	§9	`sensitivity_disp_mmf.jl` L32–79
Terminal condition, Eq. (23)	§9	`common.jl` L269
Photon number, Eq. (27)	§10.2	`verification.jl` L188–199
Fundamental soliton, Eq. (26)	§10.1	`verification.jl` L73–150
Taylor remainder test, Eq. (28)	§10.3	`verification.jl` L271–330

12 What the Tests Do and Do Not Prove

The tests prove:

- Given the model (Eq. (4)) and the parameter values (§3), the code solves the equations correctly.

- The forward propagation reproduces an exact analytical soliton to 1.1%.
- Photon number is conserved to $< 1\%$ across all tested operating points.
- The adjoint gradient has no first-order bias (Taylor slope ≈ 2).
- The cost function is computed as claimed.

The tests do not prove:

- That the physics model is appropriate for the experimental regime you care about.
- That γ, β_2, β_3 for the specific fibers in your lab match the presets in §3.
- That the optimizer finds the global minimum of Eq. (13); only that whatever local minimum it finds is computed faithfully.